

SYS 3060: Stochastic Decision Models

Assignment 4, Version 2

Rahul Sunilkumar

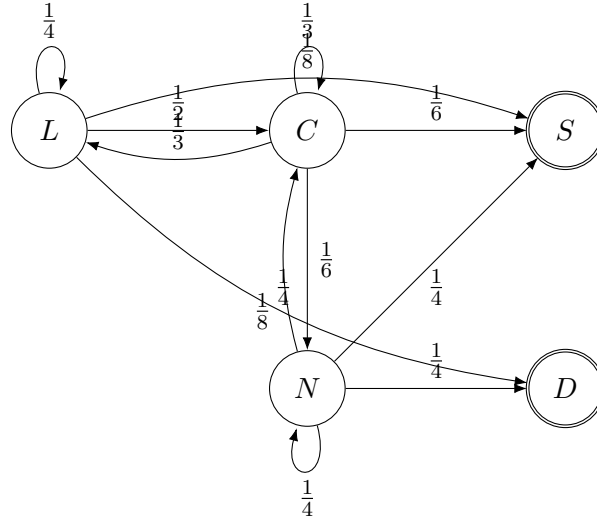
April 21, 2026

Q1. Absorbing

Use the state order

(L, C, N, S, D) ,

with transient states (L, C, N) and absorbing states (S, D) .



(a) Transition matrix in canonical form.

If we list the transient states first,

$$P = \left[\begin{array}{ccc|cc} \frac{1}{4} & \frac{1}{2} & 0 & \frac{1}{8} & \frac{1}{8} \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{6} & \frac{1}{6} & 0 \\ 0 & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \\ \hline 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{array} \right] = \begin{bmatrix} Q & R \\ 0 & I \end{bmatrix},$$

where

$$Q = \begin{bmatrix} \frac{1}{4} & \frac{1}{2} & 0 \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{6} \\ 0 & \frac{1}{4} & \frac{1}{4} \end{bmatrix}, \quad R = \begin{bmatrix} \frac{1}{8} & \frac{1}{8} \\ \frac{1}{6} & 0 \\ \frac{1}{4} & \frac{1}{4} \end{bmatrix}.$$

(b) Fundamental matrix.

$$I - Q = \begin{bmatrix} \frac{3}{4} & -\frac{1}{2} & 0 \\ -\frac{1}{3} & \frac{2}{3} & -\frac{1}{6} \\ 0 & -\frac{1}{4} & \frac{3}{4} \end{bmatrix},$$

so

$$N = (I - Q)^{-1} = \begin{bmatrix} \frac{44}{21} & \frac{12}{7} & \frac{8}{21} \\ \frac{8}{7} & \frac{18}{7} & \frac{4}{7} \\ \frac{8}{21} & \frac{6}{7} & \frac{32}{21} \end{bmatrix}.$$

(c) Expected number of evenings in each transient state before absorption, starting from L .

The first row of N gives the expected number of visits to each transient state before absorption:

$$(\mathbb{E}_L[\#L], \mathbb{E}_L[\#C], \mathbb{E}_L[\#N]) = \left(\frac{44}{21}, \frac{12}{7}, \frac{8}{21} \right).$$

Numerically,

$$\left(\frac{44}{21}, \frac{12}{7}, \frac{8}{21} \right) \approx (2.095, 1.714, 0.381).$$

(d) Absorption probabilities starting from C .

Using

$$B = (I - Q)^{-1}R = NR,$$

we get

$$B = \begin{bmatrix} \frac{9}{14} & \frac{5}{14} \\ \frac{5}{7} & \frac{2}{7} \\ \frac{4}{7} & \frac{3}{7} \end{bmatrix}.$$

Since C is the second transient state, we use the second row:

$$\mathbb{P}_C(\text{eventually absorbed in } S) = \frac{5}{7}$$

and

$$\mathbb{P}_C(\text{eventually absorbed in } D) = \frac{2}{7}.$$

(e) Short interpretation.

Starting from C , ending in S is more likely than ending in D :

$$\frac{5}{7} \approx 0.714 \quad \text{vs.} \quad \frac{2}{7} \approx 0.286.$$

So sleep is the more likely absorbing outcome here. Also, the second row of N shows that the chain tends to return to the transient states a few times before absorption, especially to C .

Q2. Workforce Planning

Let the non-leaving states be A, M, P , and let L denote leaving the lab.

(a) Balance equations.

Using the balance equation,

$$H_i + \sum_{k \neq i} N_k p_{ki} = N_i \sum_{k \neq i} p_{ik}.$$

Since new hires only enter at A , we have $H_M = H_P = 0$. Then:

For state A ,

$$H_A + N_M p_{MA} = N_A (p_{AM} + p_{AL}),$$

so

$$H_A + \frac{1}{8} N_M = \left(\frac{1}{3} + \frac{1}{6} \right) N_A = \frac{1}{2} N_A.$$

For state M ,

$$N_A p_{AM} + N_P p_{PM} = N_M (p_{MA} + p_{MP} + p_{ML}),$$

so

$$\frac{1}{3} N_A + \frac{1}{10} N_P = \left(\frac{1}{8} + \frac{1}{4} + \frac{1}{8} \right) N_M = \frac{1}{2} N_M.$$

For state P ,

$$N_M p_{MP} = N_P (p_{PM} + p_{PL}),$$

so

$$\frac{1}{4} N_M = \left(\frac{1}{10} + \frac{3}{10} \right) N_P = \frac{2}{5} N_P.$$

(b) Use the corrected target value to solve for the other staffing level.

From the P -balance equation,

$$\frac{1}{4} N_M = \frac{2}{5} N_P.$$

So the two quantities have to satisfy

$$N_P = \frac{5}{8} N_M \quad \text{or equivalently} \quad N_M = \frac{8}{5} N_P.$$

That means in the corrected version of the problem, once one of N_M or N_P is given, the other one is determined by this relation before doing the rest of the calculations.

(c) Solve for N_A .

Now use the M -balance equation,

$$\frac{1}{3}N_A + \frac{1}{10}N_P = \frac{1}{2}N_M.$$

Substituting either $N_P = \frac{5}{8}N_M$ or $N_M = \frac{8}{5}N_P$ gives

$$\frac{1}{3}N_A + \frac{1}{10}\left(\frac{5}{8}N_M\right) = \frac{1}{2}N_M$$

so

$$\frac{1}{3}N_A = \left(\frac{1}{2} - \frac{1}{16}\right)N_M = \frac{7}{16}N_M.$$

Therefore,

$$N_A = \frac{21}{16}N_M.$$

Equivalently, writing everything in terms of N_P ,

$$N_A = \frac{21}{10}N_P.$$

(d) Required average annual hiring H_A .

From the A -balance equation,

$$H_A + \frac{1}{8}N_M = \frac{1}{2}N_A.$$

Substitute $N_A = \frac{21}{16}N_M$:

$$H_A + \frac{1}{8}N_M = \frac{1}{2}\left(\frac{21}{16}N_M\right) = \frac{21}{32}N_M.$$

So

$$H_A = \left(\frac{21}{32} - \frac{1}{8}\right)N_M = \frac{17}{32}N_M.$$

Hence,

$$H_A = \frac{17}{32}N_M.$$

Using $N_M = \frac{8}{5}N_P$, this can also be written as

$$H_A = \frac{17}{20}N_P.$$

(e) Interpretation.

The hiring rate depends on whichever corrected target value the updated PDF keeps fixed. Once that target is given, the P -balance determines the other staffing level, and then N_A and H_A follow directly from the M - and A -balance equations. So the main correction is not the method; it is that we should not treat the original pair $(N_M, N_P) = (6, 4)$ as simultaneously fixed.

If the corrected PDF gives $N_M = 6$, then

$$N_P = \frac{15}{4}, \quad N_A = \frac{63}{8}, \quad H_A = \frac{51}{16}.$$

If instead it gives $N_P = 4$, then

$$N_M = \frac{32}{5}, \quad N_A = \frac{42}{5}, \quad H_A = \frac{17}{5}.$$

Q3. Knapsack

I will set this up as a finite-horizon deterministic MDP with one stage for each item type.

(a) MDP elements.

- Stages: $t = 1, 2, 3$, corresponding to items A, B, C .
- State: x_t , the remaining suitcase capacity (in kg) at the beginning of stage t .

$$x_t \in \{0, 1, \dots, 12\}, \quad x_1 = 12.$$

- Actions:

$$u_t \in \left\{ 0, 1, \dots, \left\lfloor \frac{x_t}{w_t} \right\rfloor \right\},$$

where $(w_1, w_2, w_3) = (4, 5, 7)$.

- Deterministic transition:

$$x_{t+1} = x_t - w_t u_t.$$

- Stage reward:

$$r_t(x_t, u_t) = b_t u_t,$$

where $(b_1, b_2, b_3) = (10, 12, 20)$.

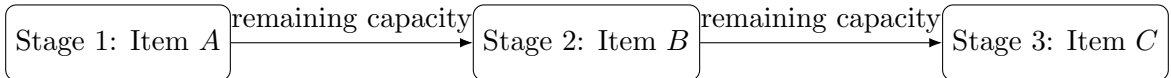
- Terminal value:

$$V_4(x) = 0 \quad \text{for all } x.$$

So the Bellman recursion is

$$V_t(x) = \max_{0 \leq u \leq \lfloor x/w_t \rfloor} \{b_t u + V_{t+1}(x - w_t u)\}, \quad t = 1, 2, 3.$$

(b) Backward induction.



Stage 3: item C only.

Since each C weighs 7 and gives benefit 20,

$$V_3(x) = \max_{0 \leq u \leq \lfloor x/7 \rfloor} 20u = \begin{cases} 0, & 0 \leq x \leq 6, \\ 20, & 7 \leq x \leq 12. \end{cases}$$

x	0	1	2	3	4	5	6	7	8	9	10	11	12
$V_3(x)$	0	0	0	0	0	0	0	20	20	20	20	20	20

Stage 2: items B and C .

$$V_2(x) = \max_{0 \leq u \leq \lfloor x/5 \rfloor} \{12u + V_3(x - 5u)\}.$$

Working this out for $x = 0, 1, \dots, 12$ gives:

x	0	1	2	3	4	5	6	7	8	9	10	11	12
$V_2(x)$	0	0	0	0	0	12	12	20	20	20	24	24	32
$u_B^*(x)$	0	0	0	0	0	1	1	0	0	0	2	2	1

In particular,

$$V_2(12) = \max\{V_3(12), 12 + V_3(7), 24 + V_3(2)\} = \max\{20, 32, 24\} = 32.$$

Stage 1: items A, B, C .

$$V_1(x) = \max_{0 \leq u \leq \lfloor x/4 \rfloor} \{10u + V_2(x - 4u)\}.$$

Doing the same thing here gives:

x	0	1	2	3	4	5	6	7	8	9	10	11	12
$V_1(x)$	0	0	0	0	10	12	12	20	20	22	24	30	32
$u_A^*(x)$	0	0	0	0	1	0	0	0	0	1	0	1	0

Starting from $x_1 = 12$,

$$u_A^*(12) = 0,$$

so we take no item A and move to stage 2 with capacity 12. Then

$$u_B^*(12) = 1,$$

so we take one item B , leaving capacity 7. Finally, at stage 3,

$$u_C^*(7) = 1.$$

So the optimal choice sequence is

$$(u_A, u_B, u_C) = (0, 1, 1).$$

(c) Optimal combination and maximum benefit.

So the best packing plan is

1 item of type B and 1 item of type C .

Its total weight is

$$5 + 7 = 12,$$

and its total benefit is

$$12 + 20 = 32.$$

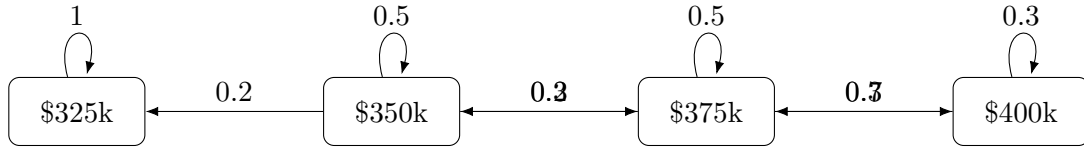
Hence

$$\boxed{\text{Maximum total benefit} = 32.}$$

Q4. MDP

Let the current best offer be measured in dollars. Since offers move in increments of \$25,000, the possible offer states are

$$325000, 350000, 375000, 400000.$$



(a) Decision epochs.

The decision epochs are the days on which an offer is observed:

$$t = 0, 1, 2, \dots$$

At each day t , after observing the current best offer, the homeowner chooses whether to accept it or wait one more day.

(b) State space.

One reasonable state space is

$$\mathcal{S} = \{325000, 350000, 375000, 400000, F\},$$

where F is a terminal “sold” state.

(c) Action space.

For any offer state $p \in \{325000, 350000, 375000, 400000\}$,

$$\mathcal{A}(p) = \{\text{accept}, \text{wait}\}.$$

For the terminal state F , no further decision is needed; we may set

$$\mathcal{A}(F) = \{\text{done}\}.$$

(d) Transition probabilities.

If the action is **accept**, then the process ends immediately:

$$\mathbb{P}(F \mid p, \text{accept}) = 1 \quad \text{for each offer state } p.$$

If the action is **wait**, then the next day's best offer follows the given transition probabilities:

$$\begin{aligned}
\mathbb{P}(325000 \mid 325000, \mathbf{wait}) &= 1, \\
\mathbb{P}(325000 \mid 350000, \mathbf{wait}) &= 0.2, \\
\mathbb{P}(350000 \mid 350000, \mathbf{wait}) &= 0.5, \\
\mathbb{P}(375000 \mid 350000, \mathbf{wait}) &= 0.3, \\
\mathbb{P}(350000 \mid 375000, \mathbf{wait}) &= 0.2, \\
\mathbb{P}(375000 \mid 375000, \mathbf{wait}) &= 0.5, \\
\mathbb{P}(400000 \mid 375000, \mathbf{wait}) &= 0.3, \\
\mathbb{P}(375000 \mid 400000, \mathbf{wait}) &= 0.7, \quad \mathbb{P}(400000 \mid 400000, \mathbf{wait}) = 0.3.
\end{aligned}$$

Finally,

$$\mathbb{P}(F \mid F, \mathbf{done}) = 1.$$

(e) Reward function and Bellman equation.

If the homeowner accepts offer p , the immediate reward is the sale price:

$$r(p, \mathbf{accept}) = p.$$

If the homeowner waits, there is no immediate payoff that day:

$$r(p, \mathbf{wait}) = 0.$$

Also,

$$r(F, \mathbf{done}) = 0.$$

Let $V(p)$ be the infinite-horizon discounted value function with discount factor $\gamma \in (0, 1)$. Then

$$V(F) = 0,$$

and for each offer state p ,

$$V(p) = \max_{a \in \mathcal{A}(p)} \left\{ r(p, a) + \gamma \sum_{q \in \mathcal{S}} \mathbb{P}(q \mid p, a) V(q) \right\}.$$

Writing these out one state at a time,

$$\boxed{V(325000) = \max \{325000, \gamma V(325000)\},}$$

$$\boxed{V(350000) = \max \{350000, \gamma (0.2V(325000) + 0.5V(350000) + 0.3V(375000))\},}$$

$$\boxed{V(375000) = \max \{375000, \gamma (0.2V(350000) + 0.5V(375000) + 0.3V(400000))\},}$$

$$\boxed{V(400000) = \max \{400000, \gamma (0.7V(375000) + 0.3V(400000))\}.}$$

These Bellman equations specify the MDP.